

SUYASH SINGH

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EDUCATION

Manipal Institute of Technology

Expected Graduation: 2027

B.Tech in Computer Science Engineering (Data Science)

Bengaluru, India

- **CGPA:** 8.51/10 | Strong foundation in CS fundamentals, Data Structures & Algorithms; full-stack & distributed systems.

TECHNICAL SKILLS

Core Fundamentals: Data Structures & Algorithms, Object-Oriented Design (OOD), System Design, Operating Systems

Languages: C++, Python, Java, TypeScript, JavaScript (ES6+), SQL

Backend & Cloud: Node.js, Express.js, FastAPI, REST APIs, Microservices, Docker, Kubernetes, NATS, KEDA, AWS, GCP, CI/CD

Observability & Data: Prometheus, Grafana, Redis, PostgreSQL, MongoDB, Qdrant Vector DB, Upstash

ML / AI & Testing: PyTorch, LLM Pipelines, RAG Frameworks, Machine Unlearning, PyTest, Jest

Competitive Programming: LeetCode (200+ Problems Solved) | CodeChef 3★ (Problem Solving & Python)

TECHNICAL PROJECTS

PathFlow "Strava for AI Agents"

Next.js 15, TypeScript, Tailwind CSS, React Flow, Prisma, OpenTelemetry, Python

- **System Architecture:** Architected an OpenTelemetry-compatible observability platform to track execution paths, token velocity (t/s), context volume, and API compute costs across autonomous agent fleets.
- **Interactive Visualization:** Built an interactive DAG visualizer with React Flow to inspect multi-step agent execution trees, sub-span latencies, and tool-routing decisions in real-time.
- **Benchmarking & Telemetry:** Engineered a multi-factor skill benchmarking engine ranking agents by micro-segment (Score = Success Rate × Efficiency × Novelty × Difficulty) and a lightweight Python SDK (@pf.trace) streaming spans to the ingestion API.

Semantic LLM Gateway & Routing Proxy

FastAPI, Qdrant, Redis, Hugging Face, Groq, Ollama

- **System Design:** Built a production-grade, AI-integrated proxy with Qdrant-backed semantic caching, drastically reducing redundant LLM API calls and achieving cache-hit latencies **under 50ms**.
- **Complex Routing:** Implemented dynamic intent routing to classify prompt complexity, intelligently distributing concurrent loads across high-performance and lightweight models to optimize resource utilization.
- **Fault Tolerance:** Engineered a robust circuit-breaker fallback mechanism, maintaining **zero-downtime** reliability and high availability during upstream rate limits or network outages.

ReachInbox Full-Stack Email Scheduler

TypeScript, Next.js, Node.js, Redis, PostgreSQL

- **Scalable Architecture:** Architected and deployed a highly concurrent email scheduling engine using Next.js and Redis, implementing distributed worker queues with exponential backoff and idempotency guarantees.
- **Full-Stack Reliability:** Engineered resilient data pipelines with PostgreSQL transactional consistency, responsive UI dashboards, and automated integration test suites for reliable message delivery.

Self-Erasing Neural Networks (SENNS) Research Publication, ICDDDS 2025

Python, PyTorch, Data Structures & Algorithms

- **Experimental Systems:** Co-authored a peer-reviewed paper designing a complex algorithmic framework for **GDPR-compliant machine unlearning**, accepted at the ICDDDS 2025 international conference.
- **Data Analysis & Visualization:** Developed rigorous diagnostic pipelines to evaluate weight shifts and per-class accuracy trade-offs, visualizing performance metrics across gradient-based data erasure models.

WORK EXPERIENCE

Turing

Dec. 2025 – May 2026

AI Intern

Bengaluru, India

- Architected and deployed scalable REST APIs and distributed inference pipelines on **AWS**, engineering highly concurrent state machine logic, third-party webhook integrations, and multi-turn session management.
- Engineered intelligent routing logic and resilient backend services with circuit breakers and automated PyTest suites, optimizing concurrent request handling and backend fault tolerance.

IEEE Computer Society Bangalore Chapter

Apr. 2025 – Sept. 2025

R&D Intern

Bengaluru, India

- Evaluated complex distributed architectures and AI research to build production-ready reference implementations.
- Analyzed system performance metrics and log data to validate architectural reliability across distributed nodes.

POSITIONS OF RESPONSIBILITY

Manipal Bengaluru Open-Source Community (MBOSC)

2024 – 2025

Project Head

- Mentored **200+ student developers** on system architecture, Git workflows, and open-source contributions, leading code reviews and encouraging software design best practices.

Codex (Competitive programming club)

2025

Project Head

- Conducted algorithmic problem-solving workshops and coding contests for **100+ members**, authoring test cases and mentoring peers in data structure optimization.